In [12]:
```
# import the libraries and and load the dataset
import pandas as pd
df = pd.read_csv(r"C:\Users\PAVAN KALYAN\Desktop\DA_projects\customer_behaviour\
```

In [2]:
```
# exploring the dataset
df.head()
```

Out[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Co
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	G
1	2	19	Male	Sweater	Clothing	64	Maine	L	Marc
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Marc
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Marc
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquo

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [4]:
```
df.describe()
```

Out[4]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

In [5]:
```
df.describe(include = 'all')
```

Out[5]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana
freq	NaN	NaN	2652	171	1737	NaN	96
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN

In [9]:
```
#checking for missing and null values in the dataset
df.isnull().sum()
```

Out[9]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [6]:
```
# filling the missing values in Review Rating column with the median rating of t
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x
```

In [14]:
```
df.isnull().sum()
```

Out[14]:
```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

In [7]:
```
#Renaming the column names for better readability and documentation

df.columns
```

Out[7]:
```
Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category',
       'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season',
       'Review Rating', 'Subscription Status', 'Shipping Type',
       'Discount Applied', 'Promo Code Used', 'Previous Purchases',
       'Payment Method', 'Frequency of Purchases'],
      dtype='object')
```

In [32]:
```
#df.columns = df.columns.str.lower()
#df.columns = df.columns.str.replace(' ','_')
#df = df.rename(columns={'Purchase Amount (USD)':'purchase_amount'})
```

```
import sqlalchemy
# Remove any parentheses, spaces, special chars
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(r'[^0-9a-zA-Z_]', '', regex=True)
df = df.rename(columns={'purchaseamountusd':'purchase_amount'})  # clean name
```

In [33]:
```
df.columns
```

Out[33]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_usd', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases', 'age_group',
       'purchase_frequency_days'],
      dtype='object')
```

In [34]:
```
#Feature engineer the dataset
labels = ['Teen', 'Adult', 'MIddle_aged', 'Senior']
df['age_group'] = pd.qcut(df['age'],q=4, labels = labels)
```

In [35]:
```
df[['age','age_group']].head(15)
```

Out[35]:

	age	age_group
0	55	MIddle_aged
1	19	Teen
2	50	MIddle_aged
3	21	Teen
4	45	MIddle_aged
5	46	MIddle_aged
6	63	Senior
7	27	Teen
8	26	Teen
9	57	MIddle_aged
10	53	MIddle_aged
11	30	Teen
12	61	Senior
13	65	Senior
14	64	Senior

In [19]:
```
#creating a new column purchase_frequency_days

df['frequency_of_purchases'].unique()
```

Out[19]:
```
array(['Fortnightly', 'Weekly', 'Annually', 'Quarterly', 'Bi-Weekly',
       'Monthly', 'Every 3 Months'], dtype=object)
```

In [36]:
```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Annually': 365,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Monthly':30,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mappi
```

In [37]:
```
df[['purchase_frequency_days','frequency_of_purchases']].head(15)
```

Out[37]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly
10	14	Bi-Weekly
11	14	Fortnightly
12	14	Fortnightly
13	7	Weekly
14	7	Weekly

In [23]:
```
df[['discount_applied','promo_code_used']].head(15)
```

Out[23]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes
10	Yes	Yes
11	Yes	Yes
12	Yes	Yes
13	Yes	Yes
14	Yes	Yes

In [38]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[38]:
```
True
```

In [33]:
```
#dropping promo_code_used column
df = df.drop('promo_code_used', axis=1)
```

In [39]:
```
df.columns
```

Out[39]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_usd', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases', 'age_group',
       'purchase_frequency_days'],
      dtype='object')
```

#connecting to the MySql database

In [26]:
```
!pip install pymysql sqlalchemy
```

```
Requirement already satisfied: pymysql in c:\users\pavan kalyan\anaconda3\lib\sit
e-packages (1.1.2)
Requirement already satisfied: sqlalchemy in c:\users\pavan kalyan\anaconda3\lib
\site-packages (2.0.34)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\pavan kalyan
\anaconda3\lib\site-packages (from sqlalchemy) (4.11.0)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\pavan kalyan\anaconda
3\lib\site-packages (from sqlalchemy) (3.0.1)
```

In [40]:
```
from sqlalchemy import create_engine

# MySQL connection
username = "root"
password = "Pavan1132.."
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{da

# Write DataFrame to MySQL
table_name = "customer"   # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

# Read back sample
pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```

Out[40]:

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	lo
0	1	55	Male	Blouse	Clothing	53	Ke
1	2	19	Male	Sweater	Clothing	64	
2	3	50	Male	Jeans	Clothing	73	Massach
3	4	21	Male	Sandals	Footwear	90	Rhode
4	5	45	Male	Blouse	Clothing	49	C

In [50]:
```
#Q1. What is the total revenue generated by male vs. female customers?

q1 = """
SELECT gender, SUM(purchase_amount_usd) AS revenue
FROM customer
GROUP BY gender;
"""
df_q1 = pd.read_sql(q1, engine)
df_q1
```

Out[50]:

	gender	revenue
0	Male	157890.0
1	Female	75191.0

In [51]:
```
#Q2. Which customers used a discount but still spent more than the average purch
q2 = """
SELECT customer_id, purchase_amount_usd
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount_usd >= (SELECT AVG(purchase_amount_usd) FROM customer);
"""
df_q2 = pd.read_sql(q2, engine)
df_q2
```

Out[51]:

	customer_id	purchase_amount_usd
0	2	64
1	3	73
2	4	90
3	7	85
4	9	97
...	...	...
834	1667	64
835	1671	73
836	1673	73
837	1674	62
838	1676	90

839 rows × 2 columns

In [49]:
```
pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```

Out[49]:

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	lo
0	1	55	Male	Blouse	Clothing	53	Ke
1	2	19	Male	Sweater	Clothing	64	
2	3	50	Male	Jeans	Clothing	73	Massach
3	4	21	Male	Sandals	Footwear	90	Rhode
4	5	45	Male	Blouse	Clothing	49	C

In [52]:
```
#Which are the top 5 products with the highest average review rating?

q3 = """
SELECT item_purchased, ROUND(AVG(review_rating), 2) AS avg_product_rating
FROM customer
GROUP BY item_purchased
ORDER BY avg_product_rating DESC
LIMIT 5;
"""
df_q3 = pd.read_sql(q3, engine)
df_q3
```

Out[52]:

	item_purchased	avg_product_rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Skirt	3.79

In [54]:

```
#Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

q4 = """
SELECT shipping_type, ROUND(AVG(purchase_amount_usd), 2) AS avg_purchase
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
"""
df_q4 = pd.read_sql(q4, engine)
df_q4
```

Out[54]:

	shipping_type	avg_purchase
0	Express	60.48
1	Standard	58.46

In [55]:

```
#Q5. Do subscribed customers spend more? Compare average spend and total revenue

q5 = """
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount_usd),2) AS avg_spend,
       ROUND(SUM(purchase_amount_usd),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue DESC, avg_spend DESC;
"""
df_q5 = pd.read_sql(q5, engine)
df_q5
```

Out[55]:

	subscription_status	total_customers	avg_spend	total_revenue
0	No	2847	59.87	170436.0
1	Yes	1053	59.49	62645.0

In [56]:

```
#Q6. Which 5 products have the highest percentage of purchases with discounts ap

q6 = """
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/C
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

```
"""
df_q6 = pd.read_sql(q6, engine)
df_q6
```

Out[56]:

	item_purchased	discount_rate
0	Hat	50.00
1	Sneakers	49.66
2	Coat	49.07
3	Sweater	48.17
4	Pants	47.37

In [57]:

```
#Q7. Segment customers into New, Returning, and Loyal based on their total numbe

q7 = """
WITH customer_type AS (
    SELECT customer_id, previous_purchases,
    CASE
        WHEN previous_purchases = 1 THEN 'New'
        WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
        ELSE 'Loyal'
    END AS customer_segment
    FROM customer
)
SELECT customer_segment, COUNT(*) AS number_of_customers
FROM customer_type
GROUP BY customer_segment;
"""
df_q7 = pd.read_sql(q7, engine)
df_q7
```

Out[57]:

	customer_segment	number_of_customers
0	Loyal	3116
1	Returning	701
2	New	83

In [58]:

```
#Q8. What are the top 3 most purchased products within each category?

q8 = """
WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)
    FROM customer
    GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
"""
```

```
df_q8 = pd.read_sql(q8, engine)
df_q8
```

Out[58]:

	item_rank	category	item_purchased	total_orders
0	1	Accessories	Jewelry	171
1	2	Accessories	Sunglasses	161
2	3	Accessories	Belt	161
3	1	Clothing	Blouse	171
4	2	Clothing	Pants	171
5	3	Clothing	Shirt	169
6	1	Footwear	Sandals	160
7	2	Footwear	Shoes	150
8	3	Footwear	Sneakers	145
9	1	Outerwear	Jacket	163
10	2	Outerwear	Coat	161

In [60]:

```
#Q9Are customers who are repeat buyers (more than 5 previous purchases) also lik


q9 = """
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
"""
df_q9 = pd.read_sql(q9, engine)
df_q9
```

Out[60]:

	subscription_status	repeat_buyers
0	Yes	958
1	No	2518

In [62]:

```
#Q10. What is the revenue contribution of each age group?

q10 = """
SELECT age_group,
       SUM(purchase_amount_usd) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
"""
df_q10 = pd.read_sql(q10, engine)
df_q10
```

Out[62]:

	age_group	total_revenue
0	Teen	62143.0
1	MIddle_aged	59197.0
2	Adult	55978.0
3	Senior	55763.0

In [ ]: